

0/20 Questions Answered

hw7-diff-props-single-mean-cheating-pre-post-AI

Student Name

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Q1

0 Points

Note: There are three options for homework 7, that differ by data topic, and you only need to complete one of the three assignments.

A study was conducted by Lee et al. (2024) to investigate changes in cheating behavior among US high school students. ChatGPT, an AI tool, was launched in November of 2022. The study had been surveying students at the same US high schools since 2019, providing an opportunity to compare cheating practices pre- and post-AI and it.

	2019	2023
Used digital device as an unauthorized aid	96	38
Did not use digital device as an unauthorized aid	1223	212

Consider the above table which depicts the number of public high school students that did and did not use a digital device as an unauthorized aid on assignments by year.

Let

- p_1 represent the proportion of public high school students in 2019 that used a digital device as an unauthorized aid in this sample
- p_2 represent the proportion of public high school students in 2019 that did not use a digital device as an unauthorized aid in this sample
- π_1 represent the corresponding population parameter for 2019
- π_2 represent the corresponding population parameter for 2023

Save Answer

Q2**5 Points**

We want to calculate a 90% CI for the difference in proportion of public high school students that used a digital device as an unauthorized aid between 2019 and 2023 (i.e. $\pi_1 - \pi_2$).

Q2.1**1 Point**

What is the point estimate for the difference in proportions? Enter only your final answer. Round to four decimal places.

[Save Answer](#)**Q2.2****1 Point**

What is the critical value for the 90% confidence interval? Enter only your final answer. Round to four decimal places. Include a picture of your work

 Please select file(s)[Select file\(s\)](#)[Save Answer](#)

Q2.3**1 Point**

What is the standard error for the confidence interval? Include a picture of your work, including the formula you used.



 Please select file(s)

Select file(s)

Save Answer

Q2.4**1 Point**

With the point estimate, critical value, and standard error we estimate the 90% CI for the difference in proportion of public high school students that used a digital device as an unauthorized aid between 2019 and 2023 to be $(-0.1184, -0.0401)$.

We will change the question slightly. Now we want to calculate a 90% CI for the difference in proportion of public high school students that used a digital device as an unauthorized aid between 2023 and 2019 (i.e. $\pi_2 - \pi_1$). Note the order of the groups has changed now.

Think about which parts of the confidence interval formula change when you change the order of the difference in proportions.

The lower bound of the new confidence interval is: (Enter only your final answer and round to four decimal places)

The upper bound of the new confidence interval is: (Enter only your final answer and round to four decimal places)

Save Answer

Q2.5**1 Point**

What do you notice about the relationship between the two confidence intervals we get when changing the order of the groups? Comment about which parts of the confidence interval formula change when you change the order of the difference in proportions.

Save Answer

Q3**6 Points**

Now we want to conduct a hypothesis test to test if there is evidence of a difference in proportions, with a significance level of 0.1.

$$H_0 : \pi_1 = \pi_2$$

$$H_A : \pi_1 \neq \pi_2$$

Q3.1**1 Point**

Calculate the pooled proportion assuming the null hypotheses is true ($H_0 : \pi_1 = \pi_2$). Enter only your final answer and round to four decimal places. Include a picture of your work.

 Please select file(s)**Select file(s)****Save Answer****Q3.2****1 Point**

Calculate the standard error for the pooled proportion. Include a picture of your work including the formula used.

 Please select file(s)**Select file(s)****Save Answer**

Q3.3**1 Point**

If the null hypothesis were true, what would be the sampling distribution of the difference in sample proportions?

Save Answer

Q3.4**1 Point**

Now we need to calculate the p-value so we can judge if there is a statistically significant difference in proportions. Enter only your final answer and round to four significant digits. (You can enter scientific notation such as 2.436×10^{-2} as 2.436e-2). Then upload a picture of your work.

 Please select file(s)

Select file(s)

Save Answer

Q3.5**2 Points**

You should have found a p-value that is less than 0.1. Select the following option that is an appropriate conclusion, at a significance level of 0.1, from the results of our hypothesis test.

We accept the null hypothesis and conclude there is no evidence of a difference in proportion of public high school students in the U.S. using a digital device as an unauthorized age between 2019 and 2023.

We fail to reject the null hypothesis and conclude there is no evidence of a difference in proportion of public high school students in the U.S. using a digital device as an unauthorized age between 2019 and 2023.

We fail to reject the null hypothesis and conclude there is evidence of a difference in proportion of public high school students in the U.S. using a digital device as an unauthorized age between 2019 and 2023.

We reject the null hypothesis and conclude there is evidence of a difference in proportion of public high school students in the U.S. using a digital device as an unauthorized age between 2019 and 2023.

We reject the null hypothesis and conclude there is no evidence of a difference in proportion of public high school students in the U.S. using a digital device as an unauthorized age between 2019 and 2023.

Save Answer

Q4**2 Points**

We forgot to check that the assumptions were met for the confidence interval and hypothesis test we performed! Please identify the conditions and explain if they were met.

Save Answer

Q5**8 Points**

Consider the following hypothetical situation:

A researcher wants to investigate the GPA of students that have cheated using an AI tool. She wants to test if the average GPA of students that cheat using AI tools is different from 3.0, at a significance level of 0.05. She surveys 100 students at on a campus that have cheated using an AI tool and gets a sample mean GPA of 2.75, with a sample standard deviation of 0.25.

Q5.1**1 Point**

The null and alternative hypothesis are:

$$H_0 : \mu = 3.0$$

$$H_A : \mu \neq 3.0$$

What is the point estimate? Enter only your final answer.

[Save Answer](#)**Q5.2****1 Point**

What is the standard error of the point estimate? Enter only your final answer. Include a picture of your work.

 Please select file(s)[Select file\(s\)](#)[Save Answer](#)

Q5.3**1 Point**

If the null hypothesis were true, what is the sampling distribution?

[Save Answer](#)**Q5.4****2 Points**

Now we need to calculate the p-value so we can judge if there is evidence of the average GPA being different than 3.0. Enter only your final answer and round to four significant digits. Then upload a picture of your work.

 Please select file(s)[Select file\(s\)](#)[Save Answer](#)

Q5.5**2 Points**

You should have calculated a p-value much smaller than the significance level, 0.05. Explain to your friend Lukas, who is a big fan of using AI tools on his assignments, what this p-value means, in context. Tell him if there is evidence, based on this data, that the average GPA of students who use AI tools to cheat differs from 3.0.

[Save Answer](#)**Q5.6****1 Point**

Say the researcher then takes another sample. With the new data of 100 students the sample mean is still 2.75 GPA, but the sample standard deviation this time is 2.5. State how the p-value would change and explain why this change makes sense.

[Save Answer](#)

Q6**0 Points**

The following two questions are related to the homework choice you made in choosing to do this assignment instead of the other two option. There are no correct answers, this question is only being graded for completeness (note that answering "I felt like it" is an insufficient response).

Please explain in detail what led you to choose this topic, cheating pre- and post-AI, to investigate.

What was it about the other two options (sleep deprivation and electric vehicle adoption) that led you not to choose them? Please explain in detail.

[Save Answer](#)

Q7**0 Points**

- Have you done the homework by yourself, with a peer(s), or used tools like ChatGPT? If you do not acknowledge any help from a person or ChatGPT, we will consider this cheating.

- If you provided help to a peer(s) and/or got help from your peers(s) please explain briefly what this entailed.

- If you used ChatGPT, include the prompts you used and explain how ChatGPT supported your learning.

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